# Assignment 5 - Clustering

*Team No : 49*

Course Name : Data Analytics - I

Course Code : CS4.405

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# 1 Data Visualization and Preprocessing Using PowerBI

This report presents an analysis of the marketing campaign dataset. The dataset contains customer information such as demographic attributes, income, and purchasing behavior. The visualizations were created in Power BI and form the basis of the insights discussed.

## 1.1 Data Overview

The dataset consists of the following key attributes:

- **Demographic Data:** Year of Birth, Education Level, Marital Status, Income.
- **Household Composition:** Number of children and teens at home.
- **Purchasing Behavior:** Amount spent on various products, number of purchases via different channels, and frequency of visits.
- **Customer Engagement:** Recency, Campaign Acceptance, and Complaint records.

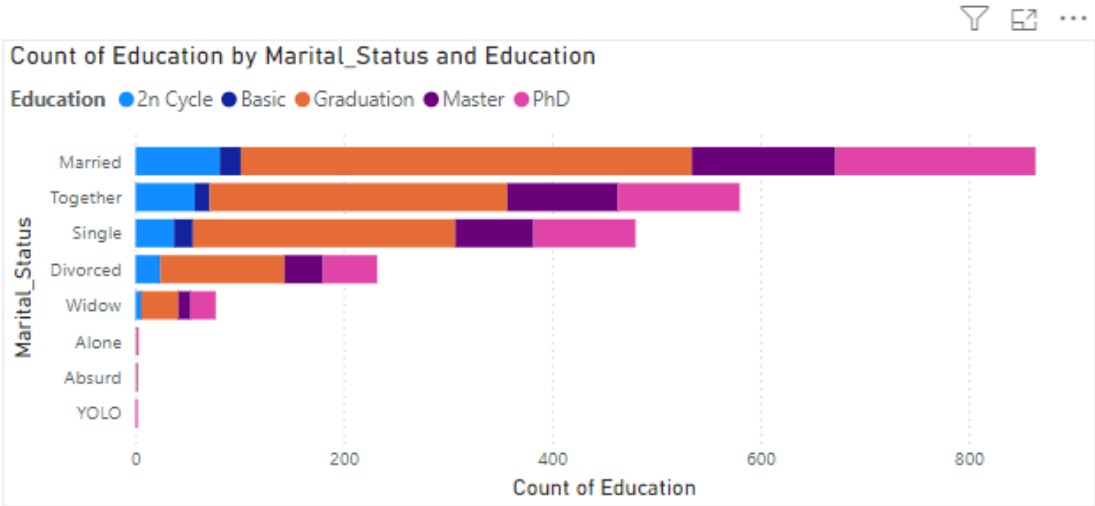
## 1.2 Bar charts to show the distribution of categorical variables

Bar charts were used to analyze the distribution of categorical variables such as *Education Level* and *Marital Status*.

### 1.2.1 Analysis of Education Levels by Marital Status

**Observations:**

- **Education Levels by Marital Status:**
  - The most significant proportion of customers across marital statuses have a **Graduation** education level.
  - **Masters** and **PhD** education levels are also prominent, especially among those who are **Married** or **Together**.
  - Lower education levels, such as **2n Cycle** and **Basic**, are less represented compared to higher education levels.
- **Marital Status Breakdown:**
  - **Married** individuals make up the largest group, with 864 customers, followed by those who are **Together** (580 customers) and **Single** (480 customers).
  - Fewer customers are **Divorced** (232), **Widow** (77), or **Alone** (3). Outliers include categories like **Absurd** and **YOLO**, with only 2 entries each.
- **Correlation Between Education and Marital Status:**
  - The **Married** group shows a wide range of education levels, with a notable concentration in **Graduation** and **PhD** categories.
  - **Together** and **Single** groups also follow a similar pattern, with most individuals in **Graduation** or higher education categories.
  - Smaller groups like **Divorced** and **Widow** still follow the trend, with **Graduation** being dominant, but they show fewer higher education qualifications like **PhD**.




| Marital_Status | Count of ID |
|---|---|
| Married | 864 |
| Together | 580 |
| Single | 480 |
| Divorced | 232 |
| Widow | 77 |
| Alone | 3 |
| Absurd | 2 |
| YOLO | 2 |
| **Total** | **2240** |

Figure 1: Distribution of Education Levels and Marital Status

- **Anomalies:**
  - Categories such as **YOLO**, **Absurd**, and **Alone** appear to be anomalies, likely representing either data entry errors or very small and unrepresentative samples.

**Insights:**

- **Target Demographics:**
  - Marketing campaigns should focus on individuals with **Graduation**, **Master**, or **PhD** education levels, as they represent the largest segments across marital statuses.
  - **Married** individuals, followed by those who are **Together**, form the core customer base and should be prioritized in tailored campaigns.
- **Educational Attainment:**
  - High education levels dominate the dataset, suggesting that the marketing strategy could appeal to individuals with a sophisticated or professional background.
- **Smaller Groups:**
  - Groups like **Widow**, **Divorced**, and **Single** could still present opportunities but require a different approach due to their smaller representation.

  - Anomalies (**YOLO**, **Absurd**) should be investigated to ensure data quality.
- **Engagement Strategy:**
  - High-education customers in the **Married** or **Together** categories are likely to respond to campaigns that align with family-oriented or professional themes.

### 1.3 Histograms to show the distribution of numerical variables

Histograms were used to analyze the distribution of numerical variables, such as *Age* and *Income*.

- The age distribution is slightly skewed towards older age groups, with a peak in the 35-50 age range.
- The income distribution is right-skewed, indicating the presence of a few high-income customers.

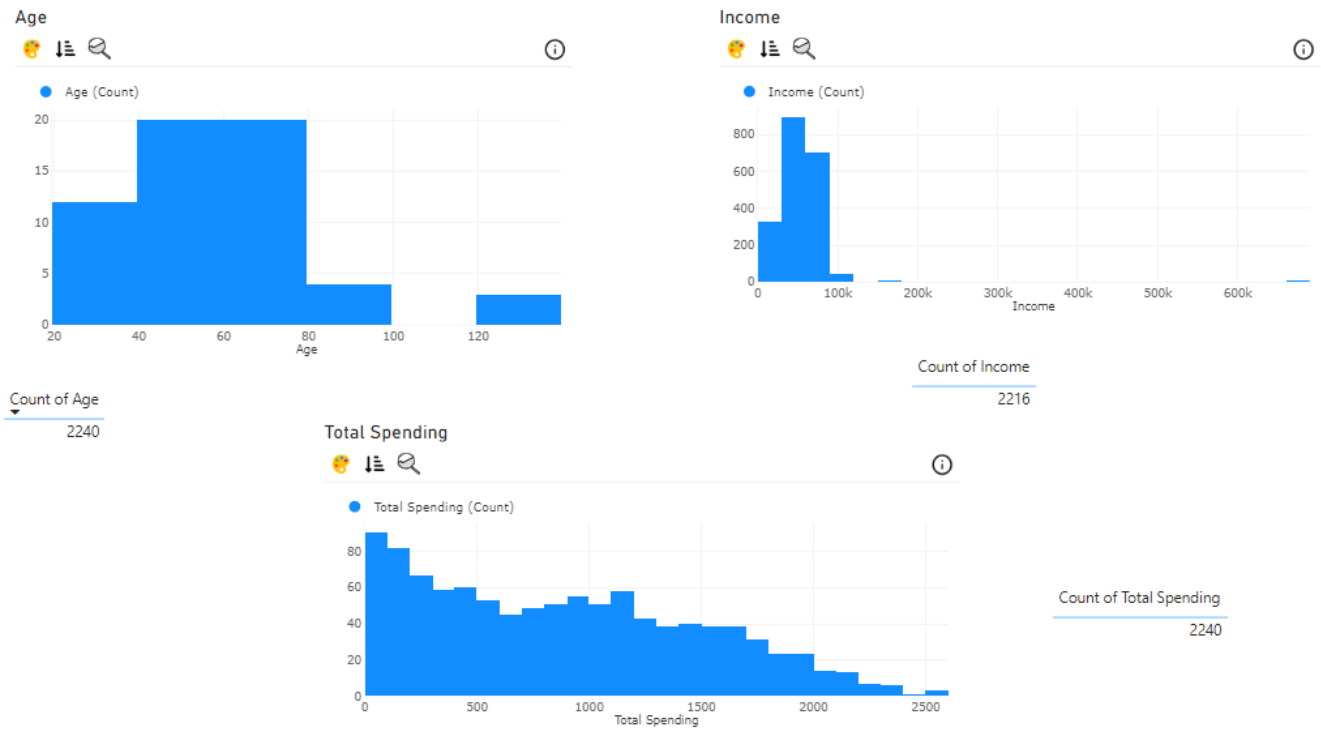



Figure 2: Distribution of Age, Income and Total spending

#### 1.3.1 Analysis of Age, Income, and Total Spending

**Observations:**

- **Age Distribution:**
  - The majority of customers fall within the age range of 30 to 60 years.
  - There are a few outliers, with ages exceeding 100 years, which could indicate potential data errors or a very small segment of elderly customers.

- **Income Distribution:**
  - Most customers have an income below Rs.100,000, with a sharp decline in counts beyond this range.
  - There are outliers with incomes exceeding Rs.200,000 and up to Rs.600,000, but these are rare occurrences.
  - The dataset contains 2,216 valid income entries, suggesting some missing values for this variable.
- **Total Spending Distribution:**
  - The distribution of total spending is right-skewed, with the majority of customers spending less than Rs.1,000.
  - Spending between Rs.1,000 and Rs.2,000 is moderately represented, while spending beyond Rs.2,000 is minimal.
  - The dataset includes all 2,240 customer records for this variable.

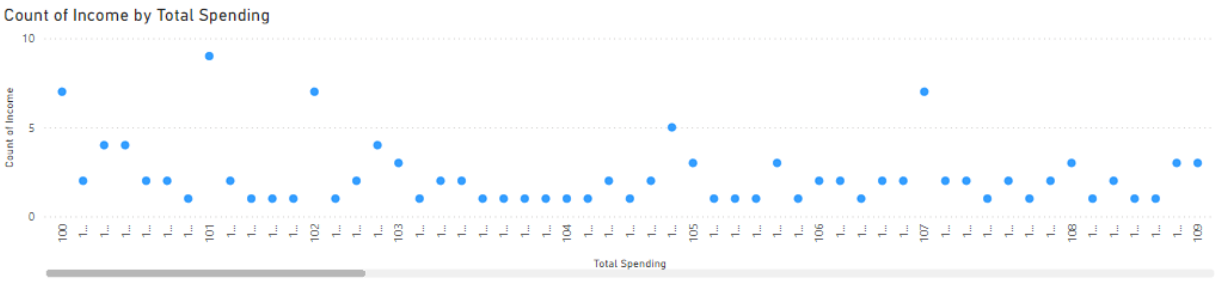
**Insights:**

- **Target Age Group:**
  - The age group of 30 to 60 years should be the focus of marketing campaigns, as it constitutes the largest customer segment.
- **Income Segmentation:**
  - Customers with an income below Rs.100,000 represent the core audience, requiring marketing strategies that align with mid-tier income brackets.
  - High-income outliers may warrant specialized campaigns targeting luxury or premium products.
- **Spending Behavior:**
  - The majority of customers are low to moderate spenders, indicating an opportunity to design promotions that incentivize higher spending.
  - High spenders, though fewer, could be targeted with loyalty programs or exclusive offers to retain and maximize value.
- **Data Quality:**
  - Outliers in age and income distributions suggest a need to review and clean the data for accuracy.
  - Missing income values should be imputed to enhance analysis and decision-making.

### 1.4 Scatter plots to identify relationships between variables

Scatter plots were used to identify relationships between variables, such as *Income* and *Total Spending*.

- A positive correlation is observed, where higher income generally aligns with higher spending.
- However, there are notable outliers, where total spending is disproportionately high or low compared to the income level.




| Total Spending | Income |
|---|---|
| 73 | 666666 |
| 108 | 162397 |
| 1761 | 160803 |
| 61 | 157733 |
| 1645 | 157243 |
| 1758 | 157146 |
| 8 | 156924 |
| 6 | 153924 |
| 304 | 113734 |
| 1754 | 105471 |
| 1141 | 102692 |
| 1266 | 102160 |
| 1162 | 101970 |

Figure 3: Relationship Between Income and Total Spending

### 1.4.1 Outliers and Clusters

- Outliers exist in both directions:
  - High spending with moderate income (e.g., spending of 1758 with an income of 157,146).
  - Low spending with high income (e.g., spending of 73 with an income of 666,666).
- Clusters of spending patterns are observed, particularly for incomes ranging between 100,000 and 160,000.

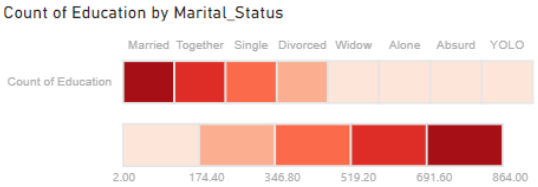
### 1.4.2 Tabular Data Insights

Key data points from the table:

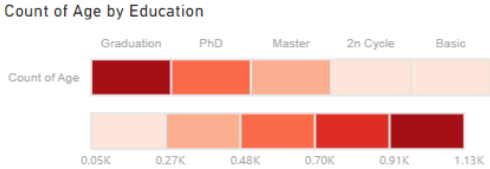
| Total Spending | Income |
|---|---|
| 73 | 666,666 |
| 108 | 162,397 |
| 1761 | 160,803 |
| 61 | 157,733 |
| 1645 | 157,243 |

## 1.5 Heatmaps or correlation matrices to identify correlations between variables

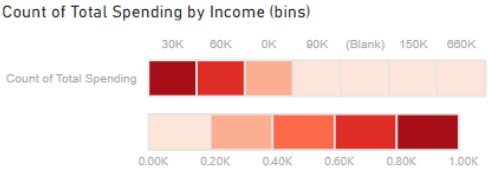
Heatmaps and correlation matrices were used to identify relationships between numerical variables.




| Education | Absurd | Alone | Divorced | Married | Single | Together | Widow | YOLO | Total |
|---|---|---|---|---|---|---|---|---|---|
| Graduation | 1 | 1 | 119 | 433 | 252 | 286 | 35 | | 1127 |
| PhD | | 1 | 52 | 192 | 98 | 117 | 24 | 2 | 486 |
| Master | 1 | 1 | 37 | 138 | 75 | 106 | 12 | | 370 |
| 2n Cycle | | | 23 | 81 | 37 | 57 | 5 | | 203 |
| Basic | | | 1 | 20 | 18 | 14 | 1 | | 54 |
| **Total** | **2** | **3** | **232** | **864** | **480** | **580** | **77** | **2** | **2240** |




| Education | 124 | 125 | 131 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Graduation | | | | | 3 | 3 | 5 | 6 | 7 | 12 | 17 | 18 | 14 | 20 | 1 |
| PhD | | 1 | | | | | | | 3 | 1 | 4 | 2 | 5 | 7 | |
| Master | | | | | | | | 2 | 2 | 1 | 3 | 2 | 4 | 7 | |
| 2n Cycle | 1 | | 1 | 1 | 2 | | | 1 | 2 | 2 | 4 | 6 | 2 | 4 | |
| Basic | | | | 1 | | | | 4 | 1 | 2 | 2 | 1 | 2 | 4 | |
| **Total** | **1** | **1** | **1** | **2** | **5** | **3** | **5** | **13** | **15** | **18** | **30** | **29** | **27** | **42** | **3** |




| Total Spending | 0 | 30000 | 60000 | 90000 | 150000 | 660000 | Total |
|---|---|---|---|---|---|---|---|
| 100 | 2 | 5 | | | | | 7 |
| 1000 | 1 | | 2 | | | | 3 |
| 1002 | | | 4 | | | | 4 |
| 1003 | | | 4 | | | | 4 |
| 1006 | | 2 | | | | | 2 |
| 1007 | | 1 | 1 | | | | 2 |
| 1008 | | | 1 | | | | 1 |
| 101 | 1 | 8 | | | | | 9 |
| **Total** | **24** | **370** | **1004** | **788** | **46** | **7** | **1** **2240** |

Figure 4: Correlation Heatmap of Numerical Variables

### 1.5.1 Count of Education by Marital Status

From the plot, it is evident that the majority of individuals with Graduation-level education fall into the Married and Together categories. Additionally:

- Single individuals predominantly have Graduation or PhD qualifications.
- There is a lower count of individuals in categories like "YOLO" or "Absurd".

### 1.5.2 Count of Age by Education

The data shows a distribution where:

- Younger individuals tend to have lower-level education (e.g., Basic or 2nd Cycle).
- Older individuals are more represented in advanced education categories such as Master's and PhD.

### 1.5.3 Count of Total Spending by Income

The total spending shows a clear trend:

- Individuals with higher income bins (e.g., 80K and above) have higher total spending.
- Lower-income bins show minimal spending, indicating potential budget constraints.

### 1.6 Key Findings

- **Customer Segmentation:** The dataset shows distinct customer groups based on age, income, and spending patterns.
- **Marketing Insights:** Married and well-educated customers form the bulk of the customer base, suggesting a targeted campaign might yield better results.
- **Potential Anomalies:** High-income customers with low spending warrant further investigation to understand their disengagement.

### 1.7 Summary

This analysis provided valuable insights into customer demographics, spending habits, and engagement levels. These findings can inform targeted marketing strategies and customer retention efforts.

## 2 K-Means Clustering

### 2.1 Implementation

- **Data Preprocessing:** We took the preprocessed data which was done by using **Power B**I tool and we further excluded the **'ID' and 'Dt_Customer'** columns from clustering and applied label encoding to categorical columns, to make it appropriate for K-Means algorithm.
- **Initialization:** Random selection of initial centroids is implemented.
- **Distance Calculation:** Euclidean distance calculation between points and centroids is correctly implemented.
- **Centroid Update:** The centroid update step is performed by averaging the points assigned to each cluster.
- **Convergence:** The algorithm stops either after a fixed number of iterations or when the centroids do not change significantly.

### 2.2 Optimal Number of Clusters

1. **Elbow Method:** We plotted the sum of squared errors (SSE) against the number of clusters. The elbow method plot shows a sharp decrease in SSE up to k = 5, indicating that k=5 might be an optimal choice as it balances within-cluster variance with the number of clusters.

2. **Silhouette Scores:** We calculated and plotted the silhouette scores for different values of k. The silhouette scores are highest for k=2, suggesting that 2 clusters provide the best separation between clusters. However, k=2 may be too broad for detailed customer segmentation.

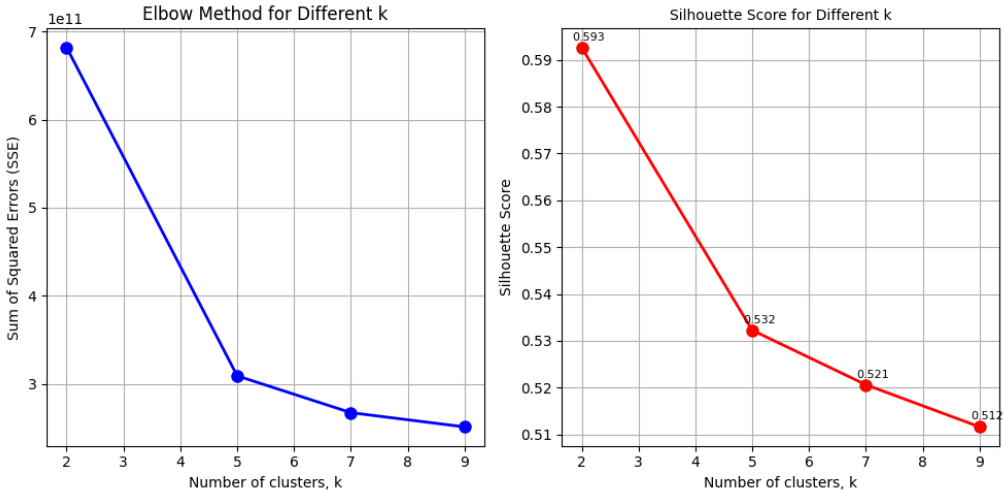



Figure 5: Number of Clusters by Elbow Method and Silhouette Scores

```
Silhouette Score for k=2: 0.5927
Silhouette Score for k=5: 0.5323
Silhouette Score for k=7: 0.5206
Silhouette Score for k=9: 0.5116
```

Figure 6: Silhouette Scores for different k

### 2.3 Analyzing Clusters Formed based on Customer Segmentation

- **K = 2 - Broad Segmentation**
  Two distinct clusters. Useful for a high-level overview.
  One cluster is densely packed, while the other has a few outliers spread out.
  Suitable for basic customer differentiation, such as separating high-value customers from low-value ones.

- **K = 5 - Detailed Segmentation**
  The plot shows five clusters with more refined groupings. There are some overlaps and outliers, but the clusters are more distinct compared to k=2.
  Provides more granularity, suitable for targeted marketing strategies.
  Identifies different customer personas and allows for personalized marketing efforts. This provides a more detailed segmentation. Each cluster could represent different customer personas, such as high-spenders, budget-conscious customers, family-oriented customers, etc.
  Based on the elbow method and silhouette scores, k=5 appears to be a balanced choice, providing detailed customer segments without over-segmentation.

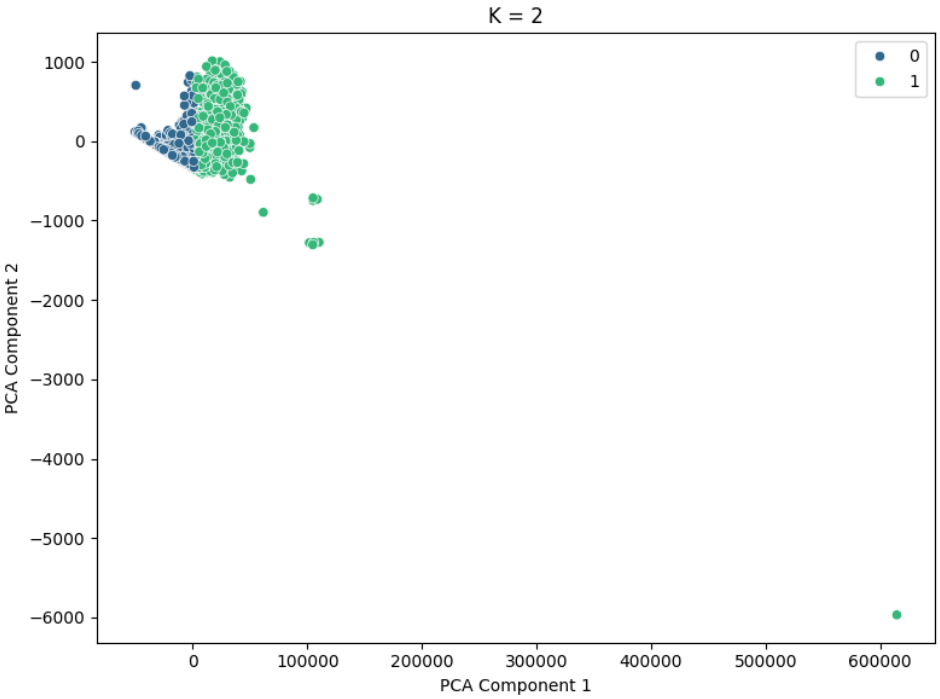



Figure 7: Scatter plot for Kmeans Clustering with k=2

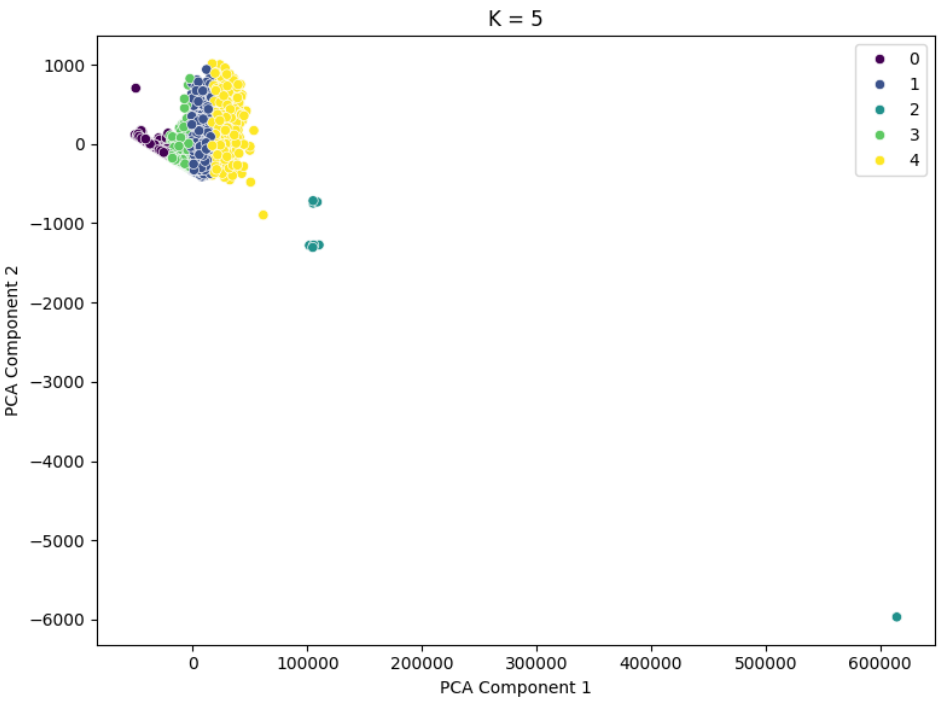



Figure 8: Scatter plot for Kmeans Clustering with k=5

- **K = 7 - Specific Segmentation**
  The plot shows seven clusters with better separation between some groups but still some overlaps and outliers.
  Offers even more detailed insights, useful for very specific marketing strategies. Helps in understanding niche customer behaviors but may have some noise.

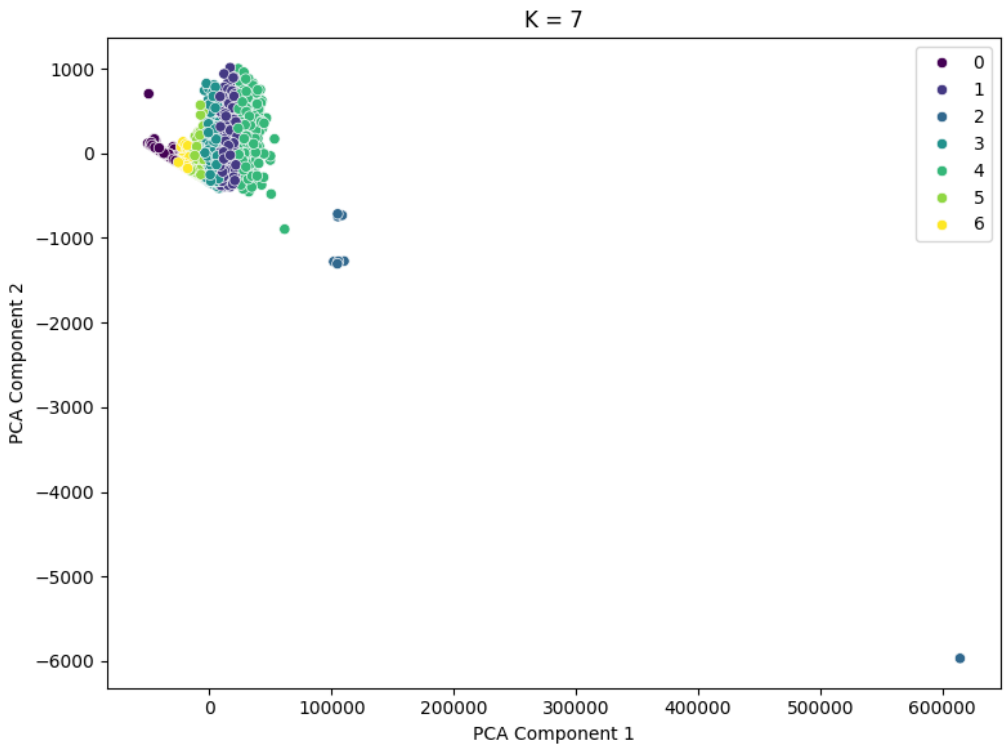



Figure 9: Scatter plot for Kmeans Clustering with k=7

- **K = 9 - Granular Segmentation**
  The plot shows nine clusters with the most granular segmentation. However, the separation is not perfect, and there are still overlaps and outliers.
  Useful for very specific customer segments but may be less practical for broad marketing strategies.

## 3 Clustering with Different Algorithms

The four CSV datasets were processed by reading the files, filling missing values with the mean, and removing rows with non-numeric data.
Scatter plots were generated for each dataset to visualize the raw data distribution.

We applied each algorithm with different number of cluster values on each dataset, and evaluated the algorithms perfromace using below evaluation metrics and compared the results.
Evaluation metrics:

- **Silhouette Score:** Measures clustering quality, excluding noise points.

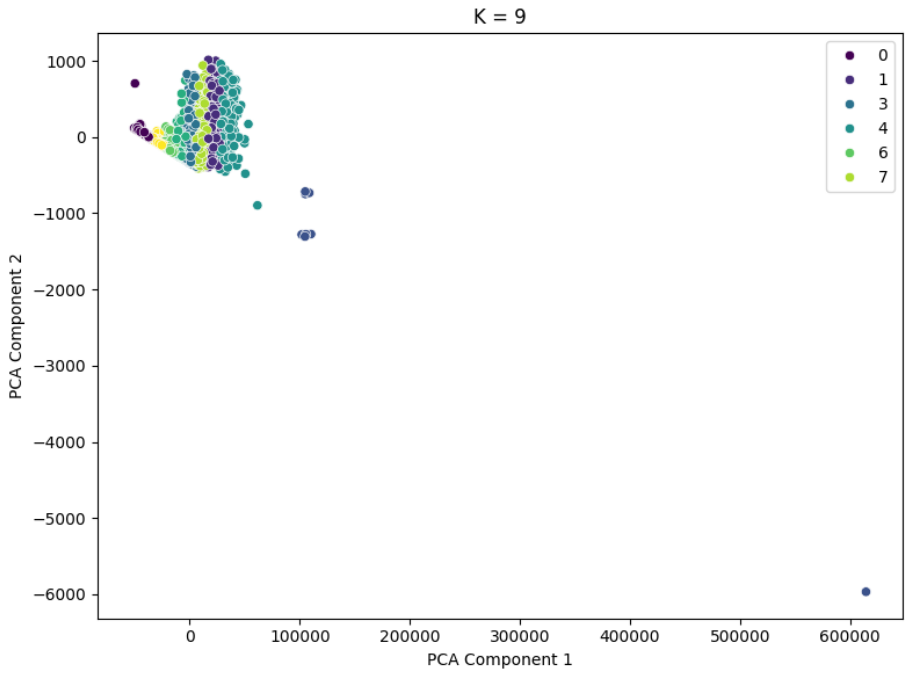



Figure 10: Scatter plot for Kmeans Clustering with k=9

- **Sum of Intra-cluster Distances:** Evaluates compactness of clusters.
- **Number of Noise Points:** Counts data points classified as noise.

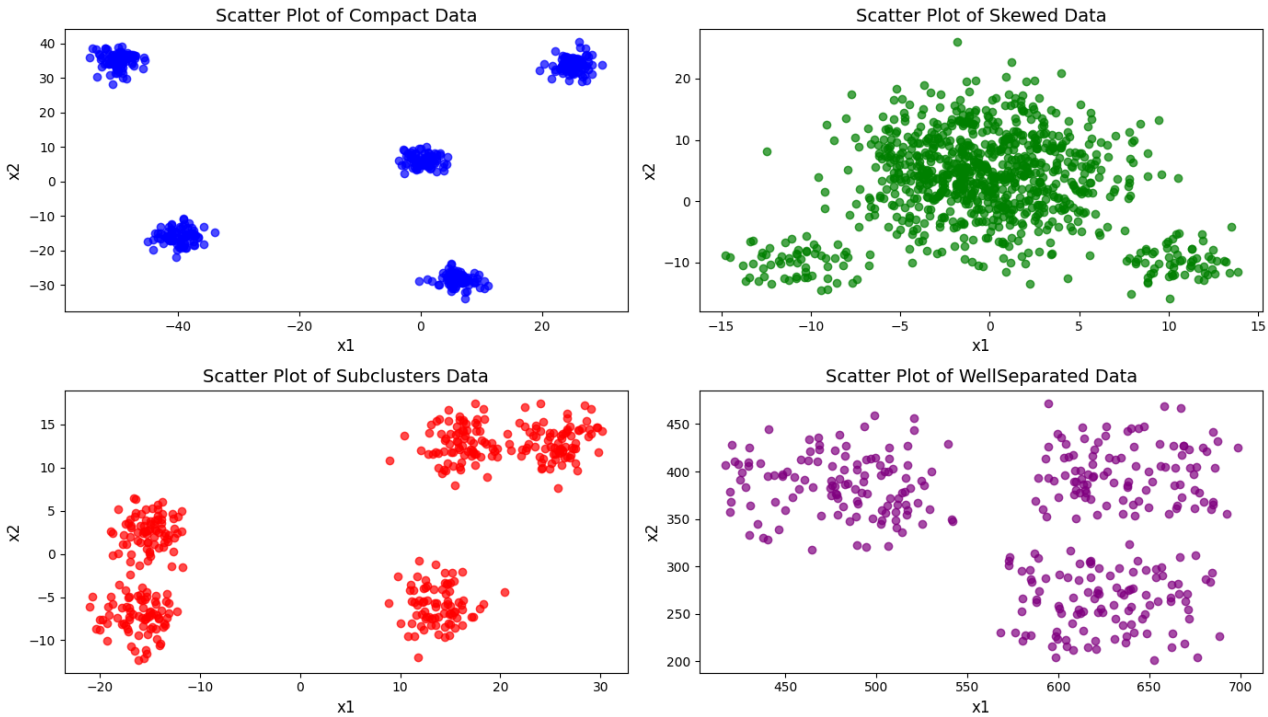



Figure 11: Raw data plots for the 4 datasets

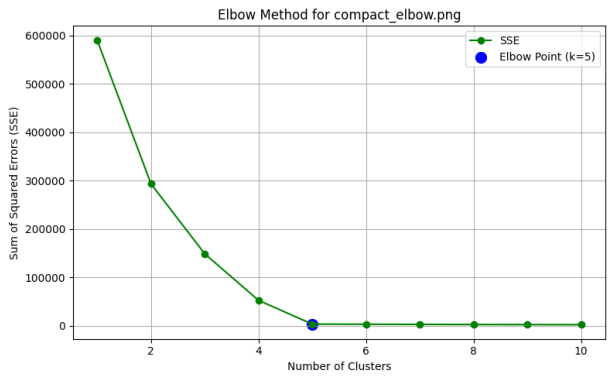
### 3.1 Kmeans Clustering and Evaluation Metrics

#### 3.1.1 Elbow Method

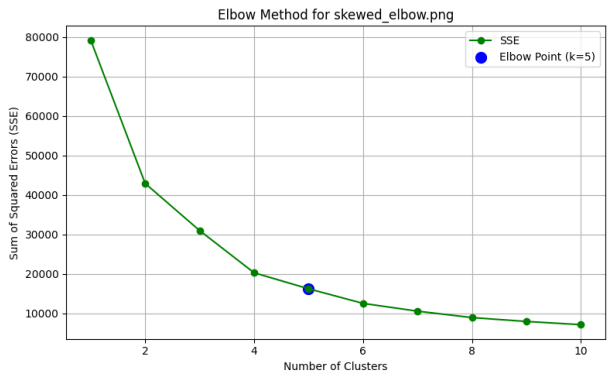
We applied KMeans clustering to each dataset by first using the Elbow method to determine the optimal number of clusters based on the Sum of Squared Errors (SSE) for each dataset.

The **optimal K results** from the elbow plots are as follows:

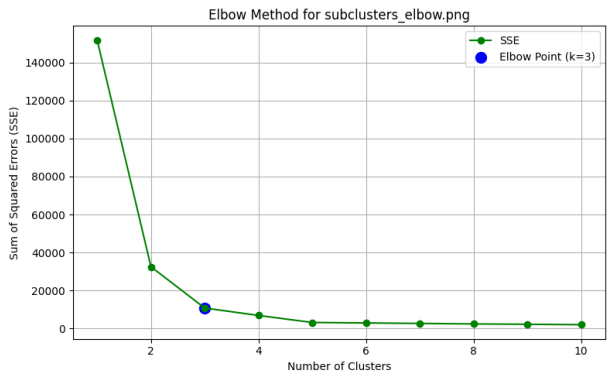
1. Compact.csv: The sharp decrease in SSE up to $K = 5$, followed by a slower decrease, suggests that the optimal number of clusters is 5.
2. Skewed.csv: The optimal number of clusters is 5.
3. Subclusters.csv: The optimal number of clusters is 3.
4. WellSeparated.csv: The optimal number of clusters is 3.



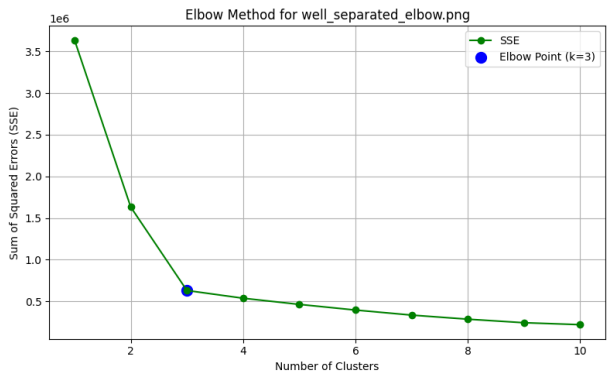

(a) Elbow Method plot for Compact.csv




(b) Elbow Method plot for Skewed.csv




(c) Elbow Method plot for Subclusters.csv




(d) Elbow Method plot for WellSeparated.csv

Figure 12: Elbow Method plots for different datasets

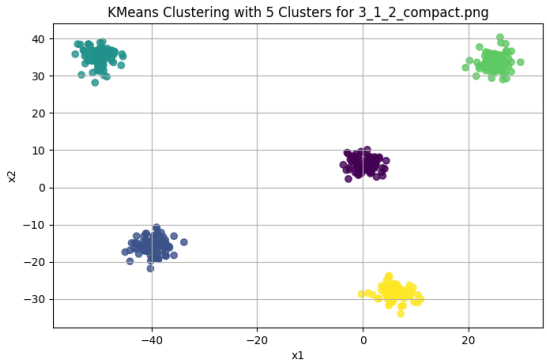
#### 3.1.2 Silhouette Score and Sum of Intra-cluster Distances

Then we applied KMeans clustering to four different datasets, using the optimal number of clusters identified through the Elbow method. For each dataset:

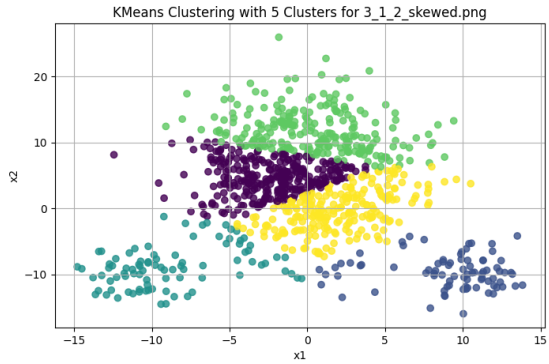
1. The silhouette score and sum of intra-cluster distances are calculated.
2. A scatter plot of the clustering result is generated and saved for each dataset.

| | Algorithm | Parameter(s) | Silhouette Score | Sum of Intra-cluster Distances |
|---|---|---|---|---|
| **kmeans_results_df1** | KMeans | n_clusters=5 | 0.91178 | 111822.247652 |
| **kmeans_results_df2** | KMeans | n_clusters=5 | 0.366567 | 1052471.416172 |
| **kmeans_results_df3** | KMeans | n_clusters=3 | 0.738949 | 372497.991882 |
| **kmeans_results_df4** | KMeans | n_clusters=3 | 0.613946 | 2063640.545711 |

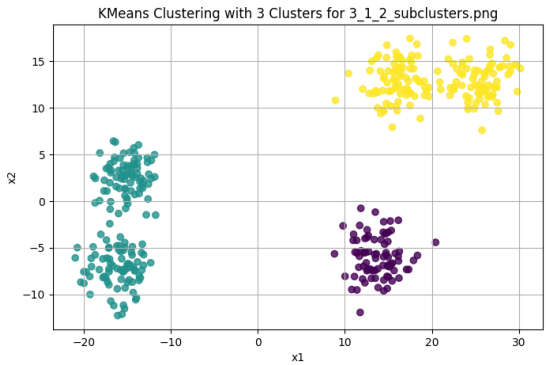
Figure 13: Kmeans Evaluation Metrics for Each Dataset taking optimal K value




(a) KMeans Clustering for Compact.csv




(b) KMeans Clustering for Skewed.csv




(c) KMeans Clustering for Subclusters.csv

(d) KMeans Clustering for WellSeparated.csv

Figure 14: KMeans Clustering plots for Each Dataset

### 3.2 Agglomerative Clustering Evaluation

For Agglomerative Clustering we applied it to four datasets using various linkage methods: **'ward', 'complete', 'average', and 'single'**. In this process:

1. Applyied Agglomerative Clustering with different values for the number of clusters ($k$) for each dataset.
2. Calculated the **silhouette score** to measure clustering quality.
3. Calculated the **sum of intra-cluster distances** to evaluate the compactness of clusters.
4. Generating scatter plots to visually represent the clustering results for each linkage method and each value of $k$.

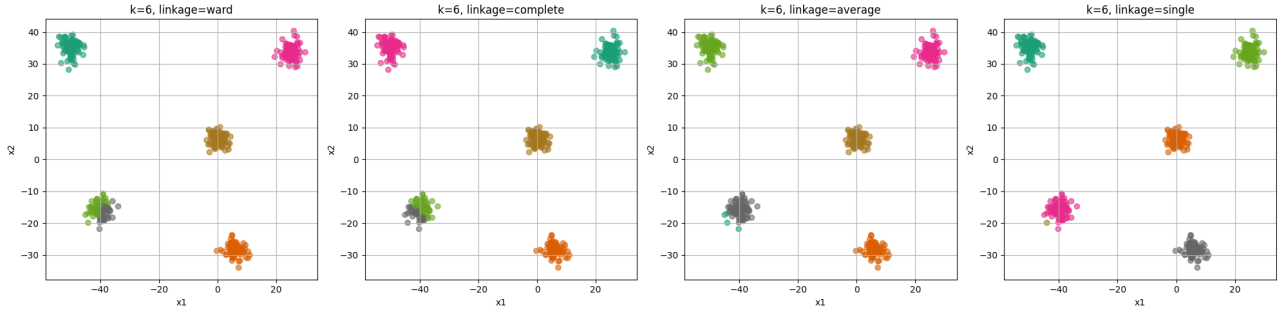



Figure 15: Agglomerative Clustering for Compact.csv with different k values

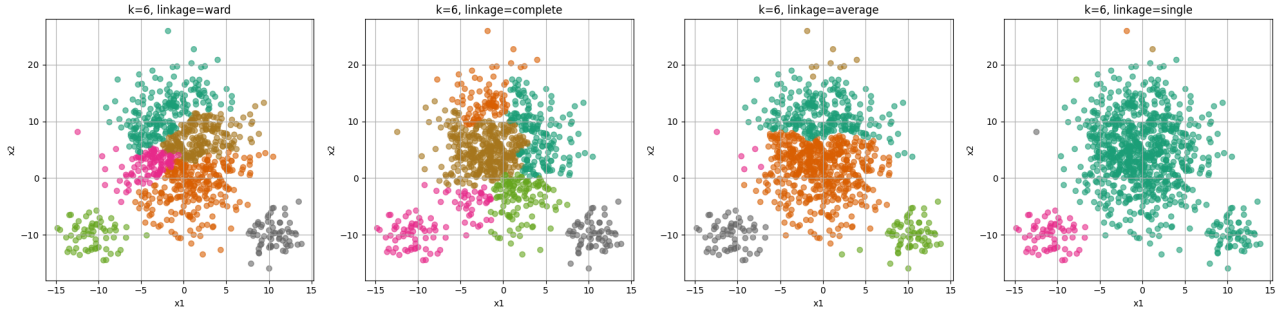



Figure 16: Agglomerative Clustering for Skewed.csv with different k values

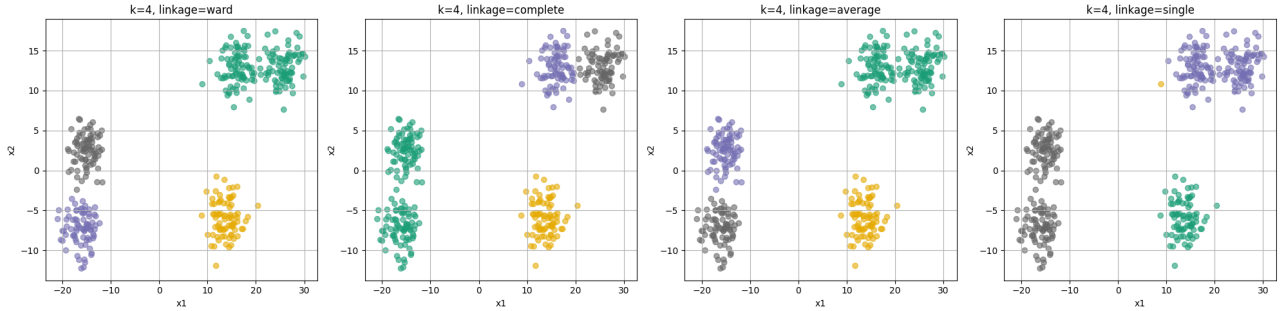



Figure 17: Agglomerative Clustering for Subclusters.csv with different k values

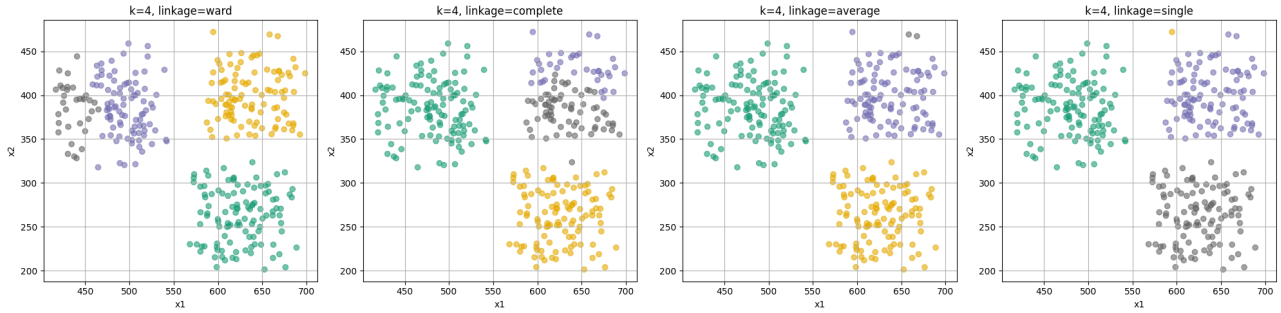



Figure 18: Agglomerative Clustering for WellSeparated.csv with different k values

```
agglo_results_df1
```

| | optimal_k | method | silhouette | sum_distances |
|---|---|---|---|---|
| 0 | 4 | ward | 0.769580 | 561995.439540 |
| 1 | 4 | complete | 0.769580 | 561995.439540 |
| 2 | 4 | average | 0.769580 | 561995.439540 |
| 3 | 4 | single | 0.769580 | 561995.439540 |
| 4 | 5 | ward | 0.911780 | 111822.247652 |
| 5 | 5 | complete | 0.911780 | 111822.247652 |
| 6 | 5 | average | 0.911780 | 111822.247652 |
| 7 | 5 | single | 0.911780 | 111822.247652 |
| 8 | 6 | ward | 0.790262 | 97449.565206 |
| 9 | 6 | complete | 0.793837 | 97335.400323 |
| 10 | 6 | average | 0.798667 | 108230.168605 |
| 11 | 6 | single | 0.797814 | 110834.819552 |

(a) Agglomerative Clustering for Compact.csv

```
agglo_results_df2
```

| | optimal_k | method | silhouette | sum_distances |
|---|---|---|---|---|
| 0 | 4 | ward | 0.391963 | 2.102685e+06 |
| 1 | 4 | complete | 0.292751 | 2.355858e+06 |
| 2 | 4 | average | 0.391412 | 2.360265e+06 |
| 3 | 4 | single | 0.173779 | 9.512837e+06 |
| 4 | 5 | ward | 0.314513 | 1.387118e+06 |
| 5 | 5 | complete | 0.312258 | 1.222168e+06 |
| 6 | 5 | average | 0.348568 | 2.311474e+06 |
| 7 | 5 | single | 0.164758 | 9.474769e+06 |
| 8 | 6 | ward | 0.304507 | 9.212792e+05 |
| 9 | 6 | complete | 0.314746 | 1.043439e+06 |
| 10 | 6 | average | 0.289924 | 2.256751e+06 |
| 11 | 6 | single | 0.197468 | 7.474048e+06 |

(b) Agglomerative Clustering for Skewed.csv

```
agglo_results_df3
```

| | optimal_k | method | silhouette | sum_distances |
|---|---|---|---|---|
| 0 | 2 | ward | 0.706530 | 906951.095639 |
| 1 | 2 | complete | 0.706530 | 906951.095639 |
| 2 | 2 | average | 0.706530 | 906951.095639 |
| 3 | 2 | single | 0.706530 | 906951.095639 |
| 4 | 3 | ward | 0.738949 | 372497.991882 |
| 5 | 3 | complete | 0.738949 | 372497.991882 |
| 6 | 3 | average | 0.738949 | 372497.991882 |
| 7 | 3 | single | 0.738949 | 372497.991882 |
| 8 | 4 | ward | 0.690470 | 241467.287308 |
| 9 | 4 | complete | 0.716513 | 244394.375844 |
| 10 | 4 | average | 0.690470 | 241467.287308 |
| 11 | 4 | single | 0.595697 | 368549.205885 |

(c) Agglomerative Clustering for Subclusters.csv

```
agglo_results_df4
```

| | optimal_k | method | silhouette | sum_distances |
|---|---|---|---|---|
| 0 | 2 | ward | 0.519528 | 5.352289e+06 |
| 1 | 2 | complete | 0.519528 | 5.352289e+06 |
| 2 | 2 | average | 0.519528 | 5.352289e+06 |
| 3 | 2 | single | 0.519528 | 5.352289e+06 |
| 4 | 3 | ward | 0.613946 | 2.063641e+06 |
| 5 | 3 | complete | 0.612632 | 2.064901e+06 |
| 6 | 3 | average | 0.613946 | 2.063641e+06 |
| 7 | 3 | single | 0.346804 | 5.286424e+06 |
| 8 | 4 | ward | 0.507298 | 1.638515e+06 |
| 9 | 4 | complete | 0.501483 | 1.740596e+06 |
| 10 | 4 | average | 0.518323 | 2.030715e+06 |
| 11 | 4 | single | 0.514372 | 2.045214e+06 |

(d) Agglomerative Clustering for WellSeparated.csv

Figure 19: Agglomerative Clustering Evaluation on Each Dataset with nearest K values of Optimal K value

### 3.3 DBSCAN Clustering Evaluation

DBSCAN was applied to the datasets using a range of `eps` values (1 to 5) and `min_samples` values (1 to 50). The process includes:

1. Clustering data points using DBSCAN for each combination of `eps` and `min_samples`.
2. Evaluated the algorithms perfromance using silhouette score, sum of intra-cluster distances, number of noise points.
3. Generated heatmaps for each metric across different parameter combinations.
4. Saving and visualizing results for each dataset to understand the impact of `eps` and `min_samples` values on clustering performance.

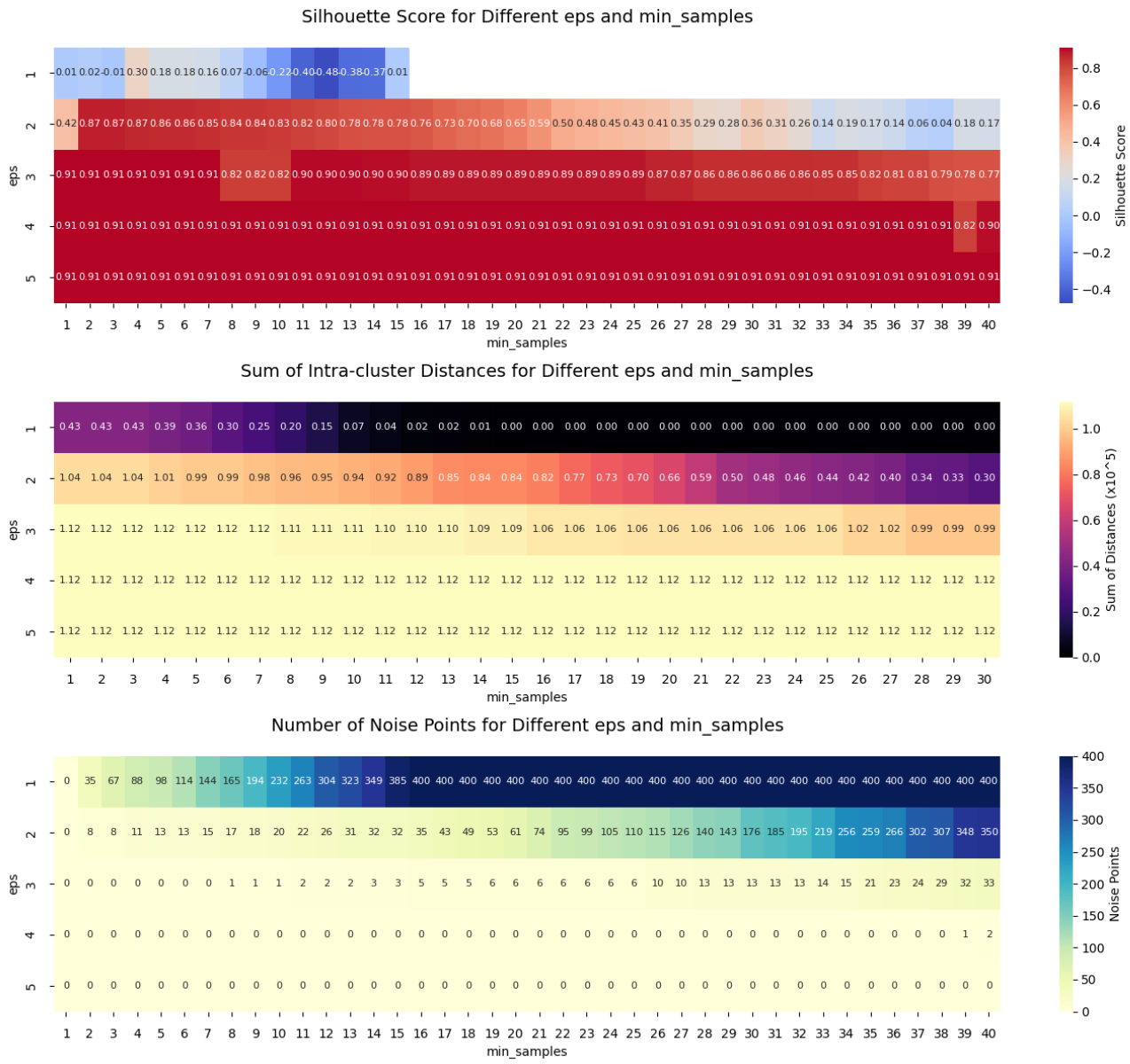



Figure 20: Heatmap for DBSCAN Clustering Results on Compact.csv

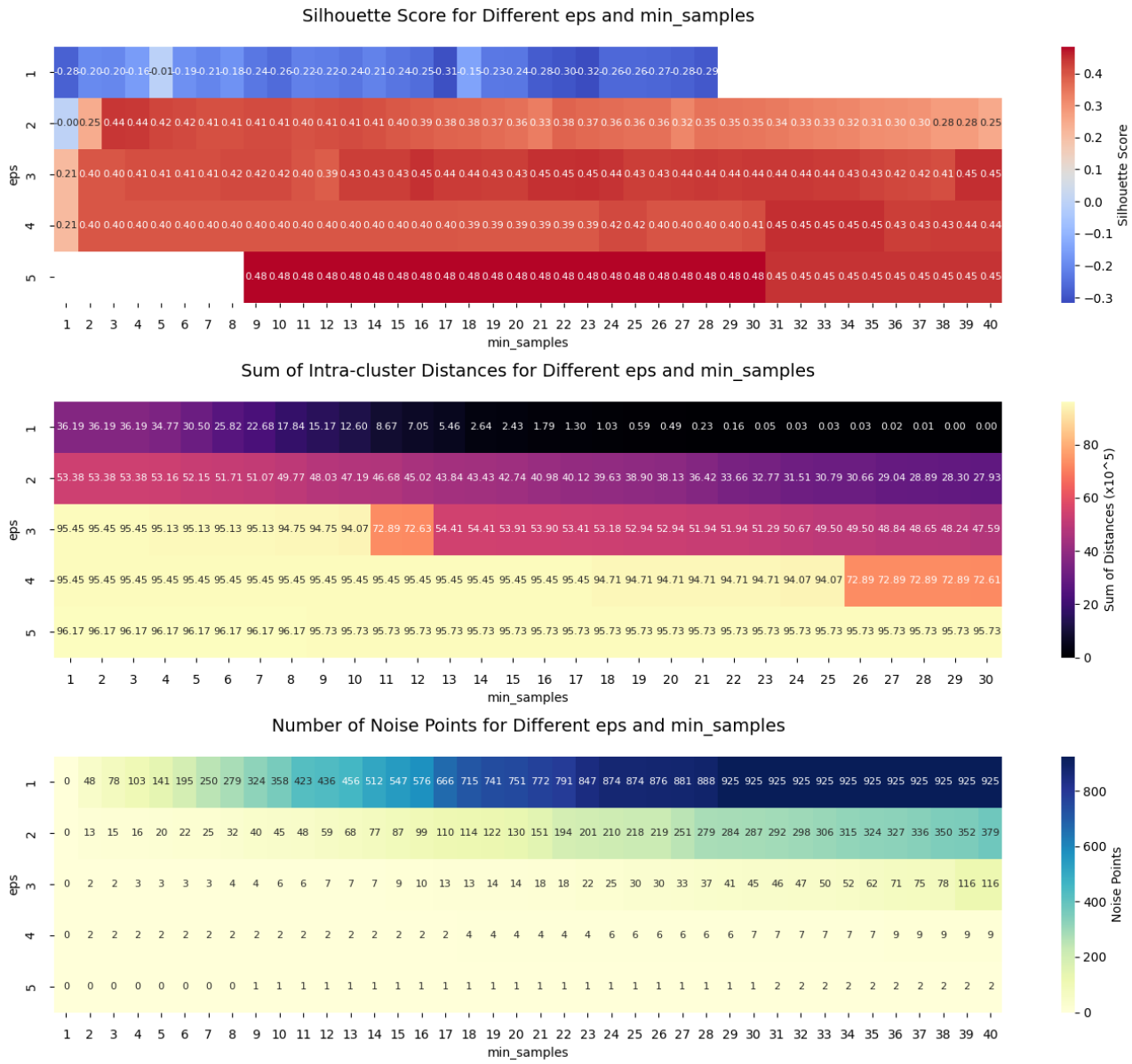



Figure 21: Heatmap for DBSCAN Clustering Results on Skewed.csv

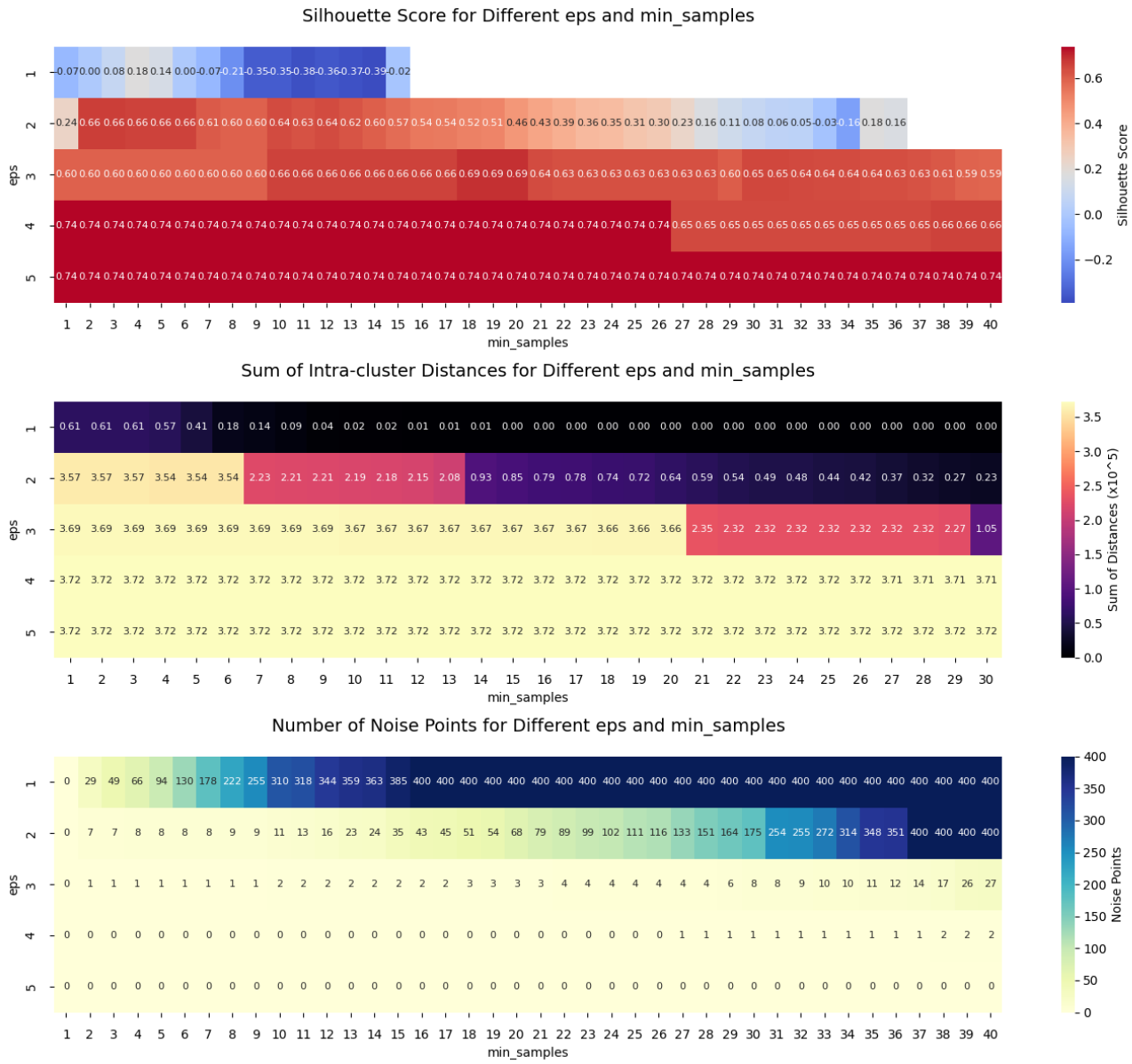



Figure 22: Heatmap for DBSCAN Clustering Results on Subclusters.csv

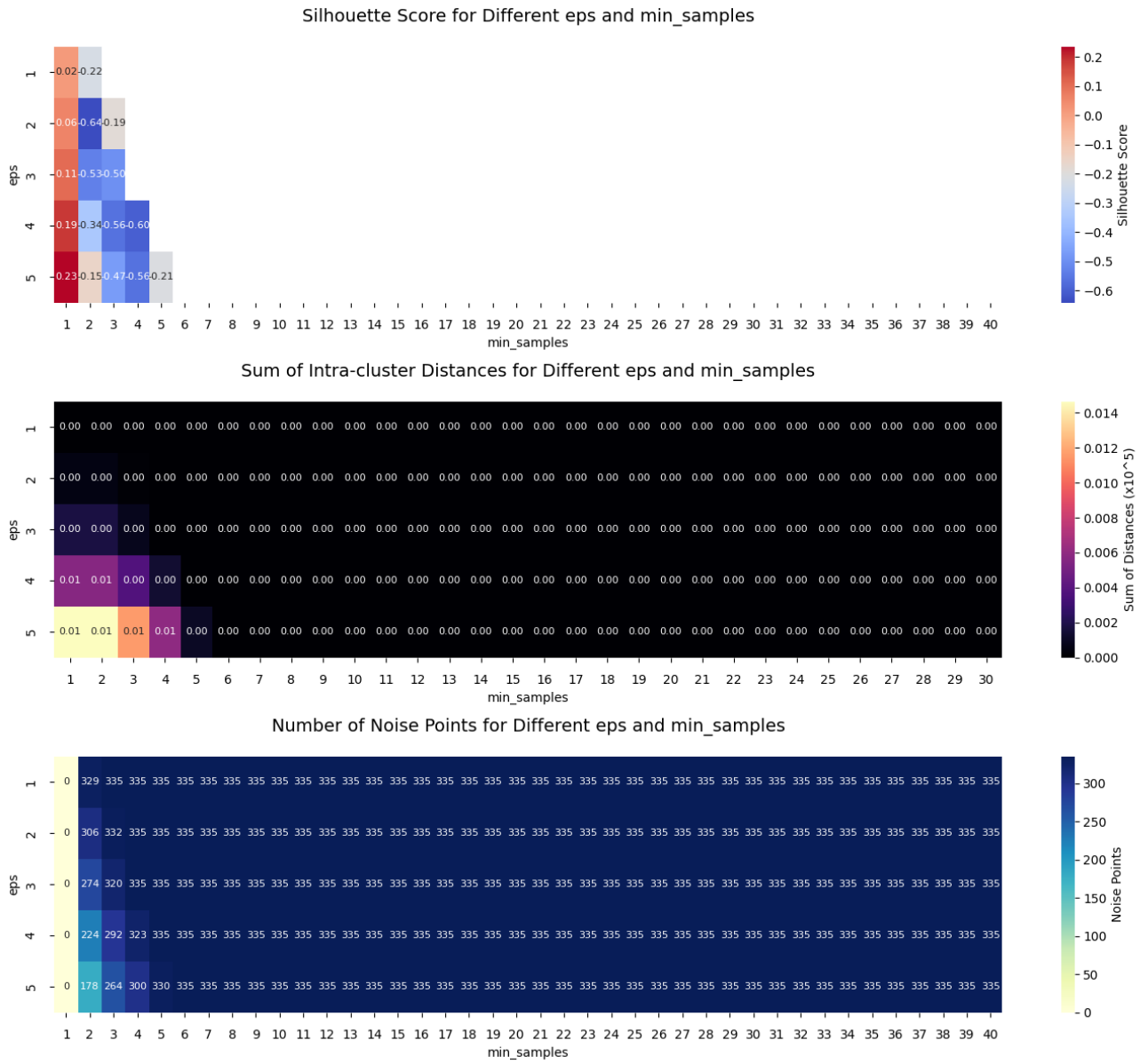



Figure 23: Heatmap for DBSCAN Clustering Results on WellSeparated.csv

dbscan_results_df1

| | eps | min_samples | silhouette | sum_distances | noise_points |
|---|---|---|---|---|---|
| **0** | 1 | 1 | 0.006413 | 42636.749258 | 0 |
| **1** | 1 | 2 | 0.024479 | 42636.749258 | 35 |
| **2** | 1 | 3 | -0.006530 | 42618.779301 | 67 |
| **3** | 1 | 4 | 0.302200 | 39237.594787 | 88 |
| **4** | 1 | 5 | 0.183287 | 36271.928597 | 98 |
| **...** | ... | ... | ... | ... | ... |
| **245** | 5 | 46 | 0.911780 | 111822.247652 | 0 |
| **246** | 5 | 47 | 0.911780 | 111822.247652 | 0 |
| **247** | 5 | 48 | 0.911780 | 111822.247652 | 0 |
| **248** | 5 | 49 | 0.911780 | 111822.247652 | 0 |
| **249** | 5 | 50 | 0.911780 | 111822.247652 | 0 |

250 rows × 5 columns

(a) DBSCAN Clustering for Compact.csv

dbscan_results_df2

| | eps | min_samples | silhouette | sum_distances | noise_points |
|---|---|---|---|---|---|
| **0** | 1 | 1 | -0.277744 | 3.618778e+06 | 0 |
| **1** | 1 | 2 | -0.199684 | 3.618778e+06 | 48 |
| **2** | 1 | 3 | -0.203755 | 3.618762e+06 | 78 |
| **3** | 1 | 4 | -0.159948 | 3.476719e+06 | 103 |
| **4** | 1 | 5 | -0.008322 | 3.049520e+06 | 141 |
| **...** | ... | ... | ... | ... | ... |
| **245** | 5 | 46 | 0.456774 | 9.502581e+06 | 3 |
| **246** | 5 | 47 | 0.456774 | 9.502581e+06 | 3 |
| **247** | 5 | 48 | 0.456774 | 9.502581e+06 | 3 |
| **248** | 5 | 49 | 0.456774 | 9.502581e+06 | 3 |
| **249** | 5 | 50 | 0.419634 | 9.474341e+06 | 4 |

250 rows × 5 columns

(b) DBSCAN Clustering for Skewed.csv

dbscan_results_df3

| | eps | min_samples | silhouette | sum_distances | noise_points |
|---|---|---|---|---|---|
| **0** | 1 | 1 | -0.069283 | 61250.804618 | 0 |
| **1** | 1 | 2 | 0.000434 | 61250.804618 | 29 |
| **2** | 1 | 3 | 0.080392 | 61239.550226 | 49 |
| **3** | 1 | 4 | 0.178685 | 57132.827923 | 66 |
| **4** | 1 | 5 | 0.141136 | 41448.974047 | 94 |
| **...** | ... | ... | ... | ... | ... |
| **245** | 5 | 46 | 0.738949 | 372497.991882 | 0 |
| **246** | 5 | 47 | 0.738949 | 372497.991882 | 0 |
| **247** | 5 | 48 | 0.738949 | 372497.991882 | 0 |
| **248** | 5 | 49 | 0.738949 | 372497.991882 | 0 |
| **249** | 5 | 50 | 0.738949 | 372497.991882 | 0 |

250 rows × 5 columns

(c) DBSCAN Clustering for Subclusters.csv

dbscan_results_df4

| | eps | min_samples | silhouette | sum_distances | noise_points |
|---|---|---|---|---|---|
| **0** | 1 | 1 | 0.01502 | 4.222569 | 0 |
| **1** | 1 | 2 | -0.22344 | 4.222569 | 329 |
| **2** | 1 | 3 | NaN | 0.000000 | 335 |
| **3** | 1 | 4 | NaN | 0.000000 | 335 |
| **4** | 1 | 5 | NaN | 0.000000 | 335 |
| **...** | ... | ... | ... | ... | ... |
| **245** | 5 | 46 | NaN | 0.000000 | 335 |
| **246** | 5 | 47 | NaN | 0.000000 | 335 |
| **247** | 5 | 48 | NaN | 0.000000 | 335 |
| **248** | 5 | 49 | NaN | 0.000000 | 335 |
| **249** | 5 | 50 | NaN | 0.000000 | 335 |

250 rows × 5 columns

(d) DBSCAN Clustering for WellSeparated.csv

Figure 24: DBSCAN Clustering Results for Each Dataset

### 3.4 Comparison and Analysis of Clustering Algorithms

We compared the performance of three clustering algorithms (KMeans, Agglomerative Clustering, and DBSCAN) across all the four datasets. The comparison is based on two key metrics: the Silhouette Score and the Sum of Intra-cluster Distances. The steps include:

1. **Determine Optimal Parameters:** For KMeans, the number of clusters (`n_clusters`) is set to 5 for datasets ‘df1‘ and ‘df2‘, and 3 for ‘df3‘ and ‘df4‘. For Agglomerative Clustering and DBSCAN, the configurations with the best Silhouette Score are selected.
2. **Results Compilation:**
   - KMeans results include Silhouette Score and Sum of Intra-cluster Distances for the optimal number of clusters.
   - Agglomerative Clustering results are derived from the best linkage method and cluster count.
   - DBSCAN results are based on the best `eps` and `min_samples` values.
3. **Visualization**: The comparison of all algorithm having the optimal results is formatted into a table, which is plotted for each dataset.

#### 3.4.1 Dataset 1: Compact.csv

- All **three algorithms** perform equally well for Compact.csv, achieving identical Silhouette Scores and Sum of Intra-cluster Distances.
- This suggests that the dataset is well-structured, allowing all methods to identify clusters effectively.

#### 3.4.2 Dataset 2: Skewed.csv

- **DBSCAN** outperforms both KMeans and Agglomerative Clustering for Skewed.csv, achieving the highest Silhouette Score.
- DBSCAN’s flexibility in handling clusters of varying densities helps it manage the skewed distribution effectively.
- KMeans and Agglomerative Clustering are less effective due to their assumptions of evenly distributed clusters.

#### 3.4.3 Dataset 3: Subclusters.csv

- All **three algorithms** perform similarly well for Subclusters.csv, achieving identical scores and distances.
- This suggests that the dataset is amenable to various clustering approaches.

#### 3.4.4 Dataset 4: WellSeparated.csv

- **KMeans and Agglomerative** Clustering outperform DBSCAN for WellSeparated.csv, achieving higher Silhouette Scores.

- DBSCAN's lower performance indicates its difficulty in dealing with this well-separated data structure.

Clustering Comparison for Compact.csv

| Algorithm | Parameter(s) | Silhouette Score | Sum of Intra-cluster Distances |
|---|---|---|---|
| KMeans | n_clusters=5 | 0.911779955803382 | 111822.24765189653 |
| Agglomerative | linkage=single, n_clusters=5 | 0.911779955803382 | 111822.24765189653 |
| DBSCAN | eps=3.0, min_samples=3.0 | 0.911779955803382 | 111822.24765189653 |

Clustering Comparison for Skewed.csv

| Algorithm | Parameter(s) | Silhouette Score | Sum of Intra-cluster Distances |
|---|---|---|---|
| KMeans | n_clusters=5 | 0.3665673143392403 | 1052471.4161720409 |
| Agglomerative | linkage=ward, n_clusters=4 | 0.3919627670943459 | 2102684.765196714 |
| DBSCAN | eps=5.0, min_samples=30.0 | 0.48364152999540017 | 9573400.831173312 |

Clustering Comparison for Subclusters.csv

| Algorithm | Parameter(s) | Silhouette Score | Sum of Intra-cluster Distances |
|---|---|---|---|
| KMeans | n_clusters=3 | 0.738949032645643 | 372497.991881627 |
| Agglomerative | linkage=single, n_clusters=3 | 0.738949032645643 | 372497.991881627 |
| DBSCAN | eps=5.0, min_samples=43.0 | 0.738949032645643 | 372497.991881627 |

Clustering Comparison for WellSeperated.csv

| Algorithm | Parameter(s) | Silhouette Score | Sum of Intra-cluster Distances |
|---|---|---|---|
| KMeans | n_clusters=3 | 0.6139458447924039 | 2063640.545711477 |
| Agglomerative | linkage=average, n_clusters=3 | 0.6139458447924039 | 2063640.545711477 |
| DBSCAN | eps=5.0, min_samples=1.0 | 0.234840705956537 | 1464.8690605278903 |

Figure 25: Comparison of three algorithms on all four datasets based on evaluation parameters

### 3.4.5 Final Result Visualization

- The following figures represent the summary plots of all three clustering algorithms (KMeans, Agglomerative Clustering, and DBSCAN) applied to each dataset with the best optimal values determined.
- Each plot shows the clusters formed by each algorithm, providing a visual comparison of their performance on the respective datasets.

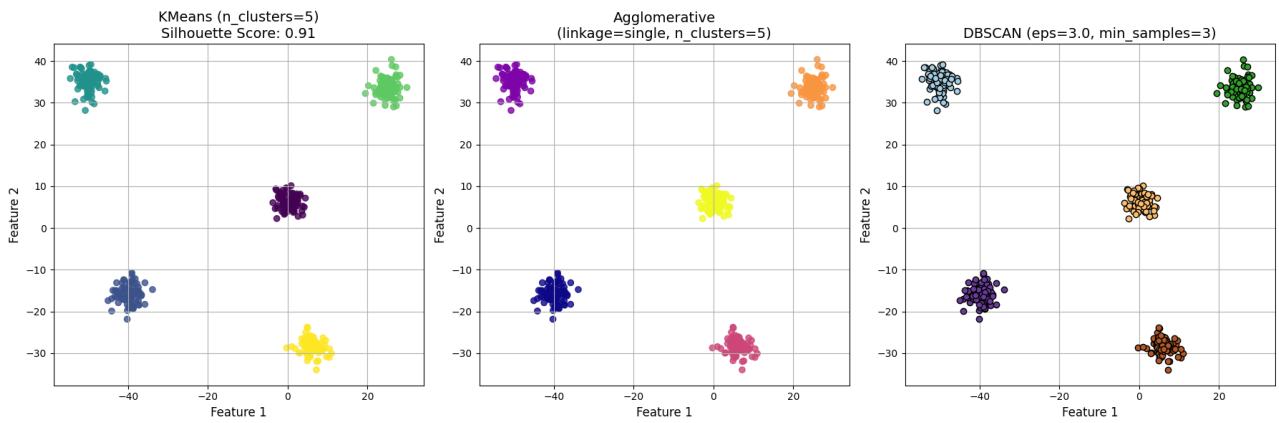



Figure 26: Optimal results for Compact.csv

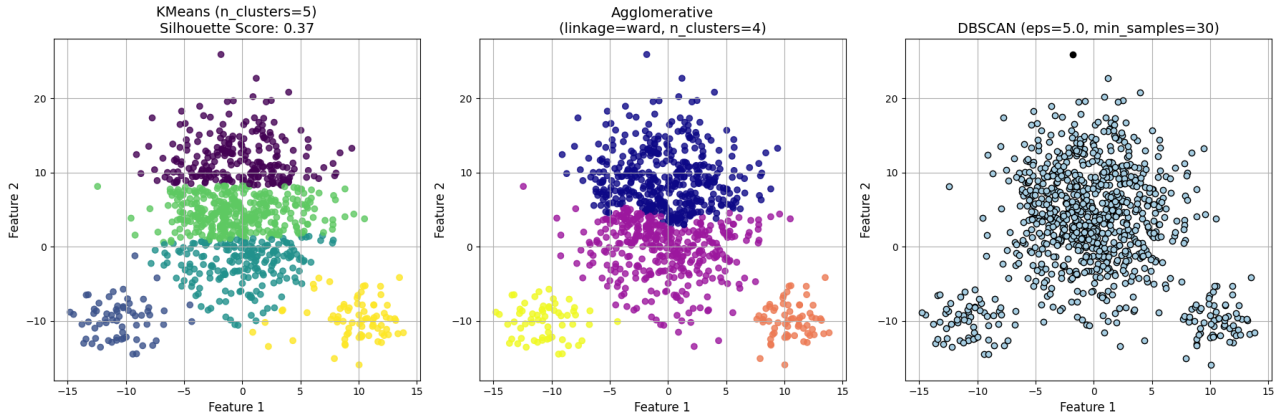



Figure 27: Optimal results for Skewed.csv

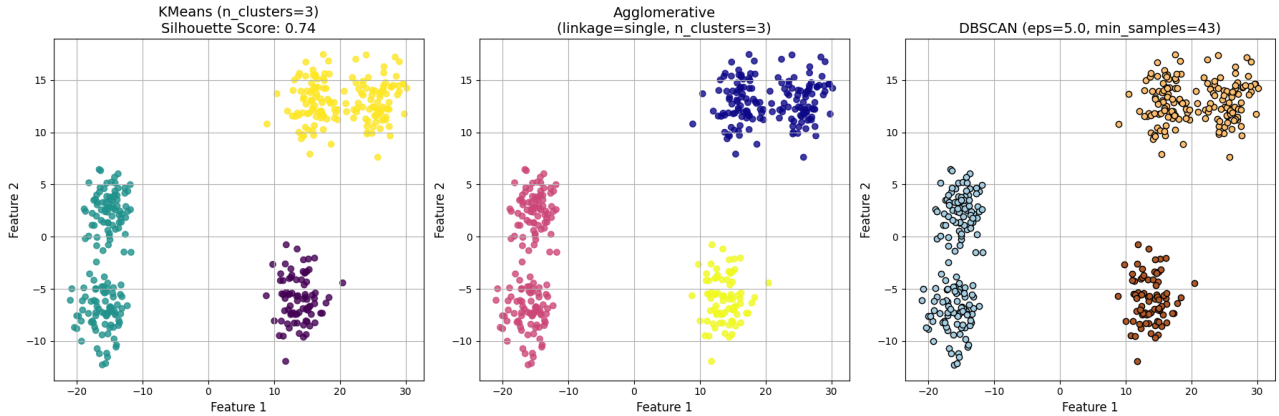



Figure 28: Optimal results for Subclusters.csv

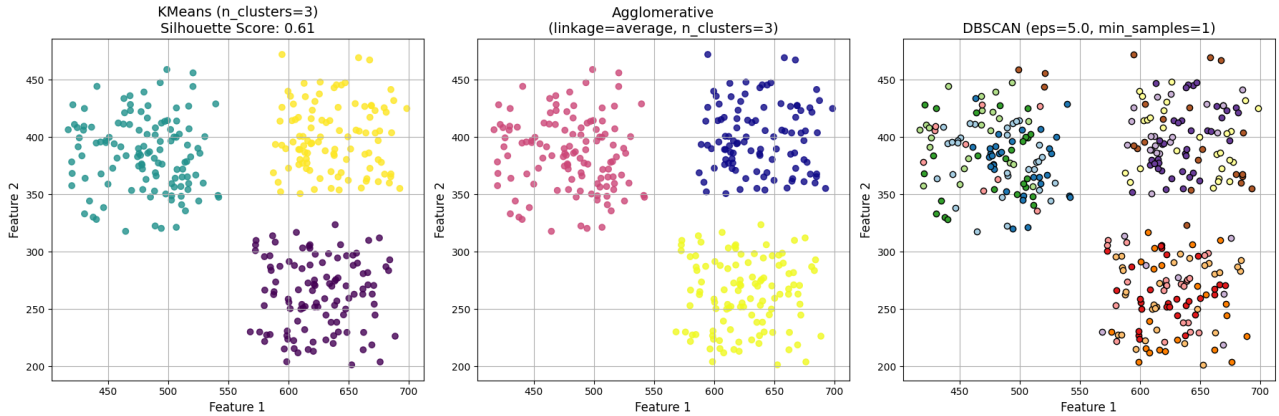



Figure 29: Optimal results for WellSeparated.csv